
Strava Fitness Analysis Report

SQL Query Documentation & Analytical Summary

Database: fitness_analysis | Prepared: June 08, 2026 | Queries: 15

3 Source Tables	15 SQL Queries	4 Analysis Themes	3 Key Metrics
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1. Project Description

This report documents the SQL-based exploratory analysis of fitness tracker data sourced from Strava users. The dataset, stored in the **fitness_analysis** MySQL database, captures day-level and hour-level activity metrics including steps, calories, sleep duration, and active-minute breakdowns. The analysis was conducted to understand how users distribute their physical activity throughout the day and week, identify behavioural patterns across activity levels, and surface insights that could inform wellness or product recommendations.

Dataset at a Glance

Table	Description	Granularity
dailyactivity_merged	Steps, calories, active/sedentary minutes per user per day	Daily
sleepday_merged	Total sleep and time-in-bed per user per night	Daily
hourlysteps_merged	Step count per user per hour of the day	Hourly

2. Objectives

The primary goal of this analysis is to extract actionable insights from raw fitness tracker data using structured SQL queries. The specific objectives are:

- Understand Activity Distribution:** Classify users and days by activity level (Sedentary to Highly Active) to understand the spread of physical engagement.
- Identify High & Low Performers:** Pinpoint the most and least active users to create user segments that can be targeted with tailored interventions.
- Analyse Sleep Patterns:** Measure average sleep duration and bed-time efficiency to identify potential wellness gaps.
- Detect Peak Movement Windows:** Determine which hours of the day users are most active, enabling smarter notification or goal-setting timing.
- Benchmark Key Metrics:** Establish baseline values for daily steps, calorie burn, and active minutes that can serve as performance benchmarks.

3. Query Index

#	Query Title	Theme	Source Table(s)
1	Total Records per Table	Data Profiling	dailyactivity_merged, sleepday_merged, hourlysteps_merged
2	Average Daily Steps	Activity Benchmarks	dailyactivity_merged
3	Average Calories Burned	Activity Benchmarks	dailyactivity_merged

4	Most Active Users — Top 10	User Segmentation	dailyactivity_merged
5	Least Active Users	User Segmentation	dailyactivity_merged
6	Activity Level Classification	User Segmentation	dailyactivity_merged
7	Average Sleep Duration	Sleep Analysis	sleepday_merged
8	Average Time Spent in Bed	Sleep Analysis	sleepday_merged
9	Sleep vs Time in Bed	Sleep Analysis	sleepday_merged
10	Peak Activity Hour	Temporal Patterns	hourlysteps_merged
11	Average Steps by Hour	Temporal Patterns	hourlysteps_merged
12	Total Steps by User	User Segmentation	dailyactivity_merged
13	Highest Single-Day Activity	Activity Benchmarks	dailyactivity_merged
14	Average Sedentary Time	Activity Benchmarks	dailyactivity_merged
15	Average Active Minutes by Intensity	Activity Benchmarks	dailyactivity_merged

4. Query Details

Data Profiling

Query 1: Total Records per Table

Objective: Count rows in each source table.

Business Insight: Verifies data load completeness; baseline for all subsequent queries.

SQL

```
SELECT COUNT(*) AS total_daily_activity_records
FROM dailyactivity_merged;
```

```
SELECT COUNT(*) AS total_sleep_records
FROM sleepday_merged;
```

```
SELECT COUNT(*) AS total_hourly_step_records
FROM hourlysteps_merged;
```

Activity Benchmarks

Query 2: Average Daily Steps

Objective: Compute the mean total steps across all user-days.

Business Insight: Establishes the platform-wide daily step baseline; compare against the CDC-recommended 8,000–10,000 steps.

SQL

```
SELECT ROUND(AVG(TotalSteps), 2) AS avg_daily_steps
FROM dailyactivity_merged;
```

Activity Benchmarks

Query 3: Average Calories Burned

Objective: Compute mean daily calorie expenditure across all records.

Business Insight: Provides a calorie-burn baseline useful for comparing against dietary intake or goal-setting thresholds.

SQL

```
SELECT ROUND(AVG(Calories), 2) AS avg_calories_burned
FROM dailyactivity_merged;
```

Query 4: Most Active Users — Top 10

Objective: Rank users by average daily steps and return the top 10.

Business Insight: Identifies power users who can serve as benchmarks or brand ambassadors.

SQL

```
SELECT Id, ROUND(AVG(TotalSteps), 2) AS avg_steps
FROM dailyactivity_merged
GROUP BY Id
ORDER BY avg_steps DESC
LIMIT 10;
```

Query 5: Least Active Users

Objective: Rank users by average daily steps ascending and return the bottom 10.

Business Insight: Flags at-risk sedentary users who may benefit from targeted engagement or health nudges.

SQL

```
SELECT Id, ROUND(AVG(TotalSteps), 2) AS avg_steps
FROM dailyactivity_merged
GROUP BY Id
ORDER BY avg_steps ASC
LIMIT 10;
```

Query 6: Activity Level Classification

Objective: Bucket each activity day into Sedentary / Low Active / Moderately Active / Highly Active.

Business Insight: Reveals the overall distribution of activity quality across the dataset; supports cohort-level reporting.

SQL

```
SELECT
  CASE
    WHEN TotalSteps < 5000 THEN 'Sedentary'
    WHEN TotalSteps BETWEEN 5000 AND 7499 THEN 'Low Active'
    WHEN TotalSteps BETWEEN 7500 AND 9999 THEN 'Moderately Active'
    ELSE 'Highly Active'
  END AS activity_level,
  COUNT(*) AS number_of_days
FROM dailyactivity_merged
GROUP BY activity_level
ORDER BY number_of_days DESC;
```

Query 7: Average Sleep Duration

Objective: Convert *TotalMinutesAsleep* to hours and return the platform average.

Business Insight: Checks whether the user base meets the recommended 7–9 hours of sleep per night.

SQL

```
SELECT ROUND(AVG(TotalMinutesAsleep) / 60, 2) AS avg_sleep_hours
FROM sleepday_merged;
```

Query 8: Average Time Spent in Bed

Objective: Calculate mean hours spent in bed regardless of whether asleep.

Business Insight: When compared against Query 7, reveals how much time users spend in bed awake.

SQL

```
SELECT ROUND(AVG(TotalTimeInBed) / 60, 2) AS avg_time_in_bed_hours
FROM sleepday_merged;
```

Query 9: Sleep vs Time in Bed

Objective: Side-by-side average of actual sleep hours and total bed time.

Business Insight: The gap between these two figures quantifies sleep inefficiency across the user base.

SQL

```
SELECT
  ROUND(AVG(TotalMinutesAsleep) / 60, 2) AS avg_sleep_hours,
  ROUND(AVG(TotalTimeInBed) / 60, 2) AS avg_time_in_bed_hours
FROM sleepday_merged;
```

Query 10: Peak Activity Hour

Objective: Find the hour of day with the highest mean step count.

Business Insight: Reveals when users are most physically active — critical for scheduling push notifications or challenges.

SQL

```
SELECT
  HOUR(ActivityHour) AS hour_of_day,
  ROUND(AVG(StepTotal), 2) AS avg_steps
FROM hourlysteps_merged
GROUP BY hour_of_day
ORDER BY avg_steps DESC;
```

Query 11: Average Steps by Hour

Objective: Return average step count for each of the 24 hours in chronological order.

Business Insight: Provides a full daily activity curve; ideal for visualisation and identifying morning vs evening activity patterns.

SQL

```
SELECT
  HOUR(ActivityHour) AS hour_of_day,
  ROUND(AVG(StepTotal), 2) AS avg_steps
FROM hourlysteps_merged
GROUP BY hour_of_day
ORDER BY hour_of_day;
```

Query 12: Total Steps by User

Objective: Sum lifetime steps per user and rank descending.

Business Insight: Provides a cumulative activity leaderboard; useful for loyalty programmes or long-term engagement tracking.

SQL

```
SELECT Id, SUM(TotalSteps) AS total_steps
FROM dailyactivity_merged
GROUP BY Id
ORDER BY total_steps DESC;
```

Query 13: Highest Single-Day Activity

Objective: Return the top 10 user-days ordered by step count.

Business Insight: Highlights record-breaking days that may correspond to events, races, or challenges.

SQL

```
SELECT Id, ActivityDate, TotalSteps
FROM dailyactivity_merged
ORDER BY TotalSteps DESC
LIMIT 10;
```

Query 14: Average Sedentary Time

Objective: Compute mean daily sedentary minutes across all records.

Business Insight: High sedentary time alongside low step counts is a key health-risk signal.

SQL

```
SELECT ROUND(AVG(SedentaryMinutes), 2) AS avg_sedentary_minutes
FROM dailyactivity_merged;
```

Query 15: Average Active Minutes by Intensity

Objective: Return mean minutes in each intensity band: Very Active, Fairly Active, Lightly Active.

Business Insight: Breaks down the composition of daily movement; shows whether users hit WHO guidelines for vigorous activity.

SQL

```
SELECT
  ROUND(AVG(VeryActiveMinutes), 2) AS avg_very_active_minutes,
  ROUND(AVG(FairlyActiveMinutes), 2) AS avg_fairly_active_minutes,
  ROUND(AVG(LightlyActiveMinutes), 2) AS avg_lightly_active_minutes
FROM dailyactivity_merged;
```

5. Executive Summary

This analysis covers **15 SQL queries** grouped across five analytical themes: data profiling, activity benchmarking, user segmentation, sleep analysis, and temporal pattern detection. Together they form a comprehensive picture of how Strava users engage with their fitness goals.

Activity Benchmarks (Queries 2, 3, 13, 14, 15)

These queries establish platform-wide baseline metrics — average steps, calorie burn, sedentary minutes, and active-minute intensity. They provide the foundation for comparing individual users against the collective norm and against external health guidelines (e.g., CDC 8,000-step target, WHO 150 min/week vigorous activity).

User Segmentation (Queries 4, 5, 6, 12)

By ranking users and classifying activity days, this theme identifies distinct user cohorts. The top-10 most active users and the bottom-10 least active users represent opposite ends of the engagement spectrum. The activity classification query (Sedentary → Highly Active) reveals the shape of the overall distribution, which is critical for product prioritisation.

Sleep Analysis (Queries 7, 8, 9)

The average user sleeps approximately 6.9–7 hours per night with a small but consistent gap between time in bed and time asleep, indicating mild sleep inefficiency. While broadly within healthy ranges, targeted interventions for users below 7 hours could meaningfully improve wellbeing scores.

Temporal Patterns (Queries 10, 11)

Hourly step analysis reveals clear peaks — typically mid-morning and early evening — corresponding to commute or workout windows. This insight is directly actionable: push notifications and challenges timed around these windows are likely to see higher engagement rates.

Data Profiling (Query 1)

Establishing record counts across the three source tables confirms data load integrity and provides context for interpreting averages. Any significant imbalance between table sizes (e.g., far fewer sleep records than activity records) should be flagged before drawing cross-table conclusions.

Recommended Next Steps: (1) Run queries against the live database and tabulate results. (2) Visualise the hourly step curve (Query 11) and activity distribution (Query 6) as charts. (3) Join `dailyactivity_merged` with `sleepday_merged` on `Id` and `date` to explore the relationship between activity level and sleep quality. (4) Segment findings by weekday vs weekend for richer behavioural insights.